Computer Vision Lecture 4
Image Segmentation – Part B

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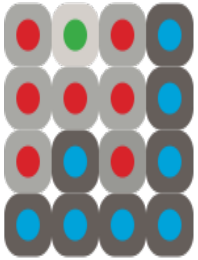
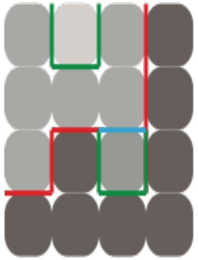
Segmentation strategies



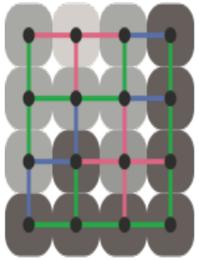
<u>Clustering in Feature Space</u>

Key idea: group pixels that have similar image features then use refinement strategies



<u>Variational Methods</u>

Key idea: define image regions based on discontinuities in the image.



<u>Graph Based Methods</u>

Key idea: model the relationships between pixels.

Segmentation by graph-theoretic clustering

- We will use the affinity or distance measures to define a graph
- This graph will then be cut to obtain “good” pieces
- The graph will be cut into connected components that have relatively large interior weights by cutting edges with low weights
- The resulting connected components will be the segmented regions

Affinity measures

We can measure the affinity of pixels, superpixels or segments

Affinity by distance:

$$\mathrm{aff}(x, y) = exp\{-((x-y)^t(x-y)/2\sigma_d^2\}$$

Affinity by intensity:

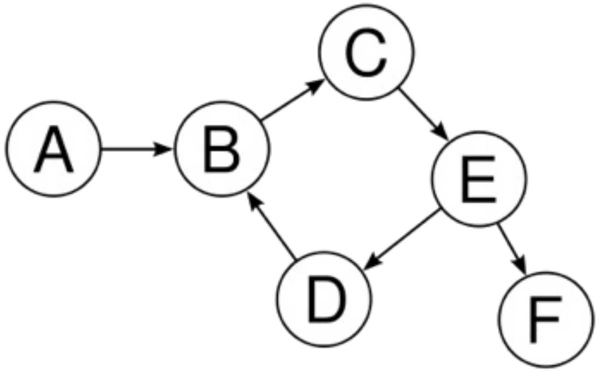
$$\mathrm{aff}(x, y) = exp\{-((I(x)-I(y))^t(I(x)-I(y))/2\sigma_I^2\}$$

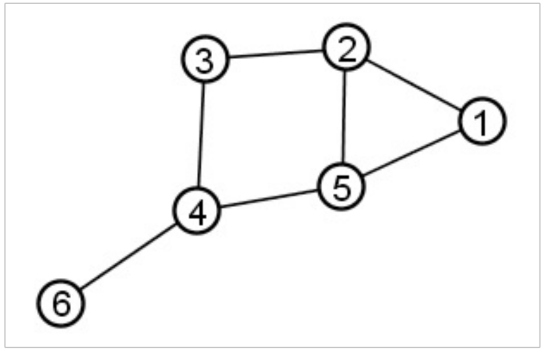
Affinity measures

We could define similar affinity measures to capture:

- Colour differences
- Texture features
- Motion (in case we have temporal data)

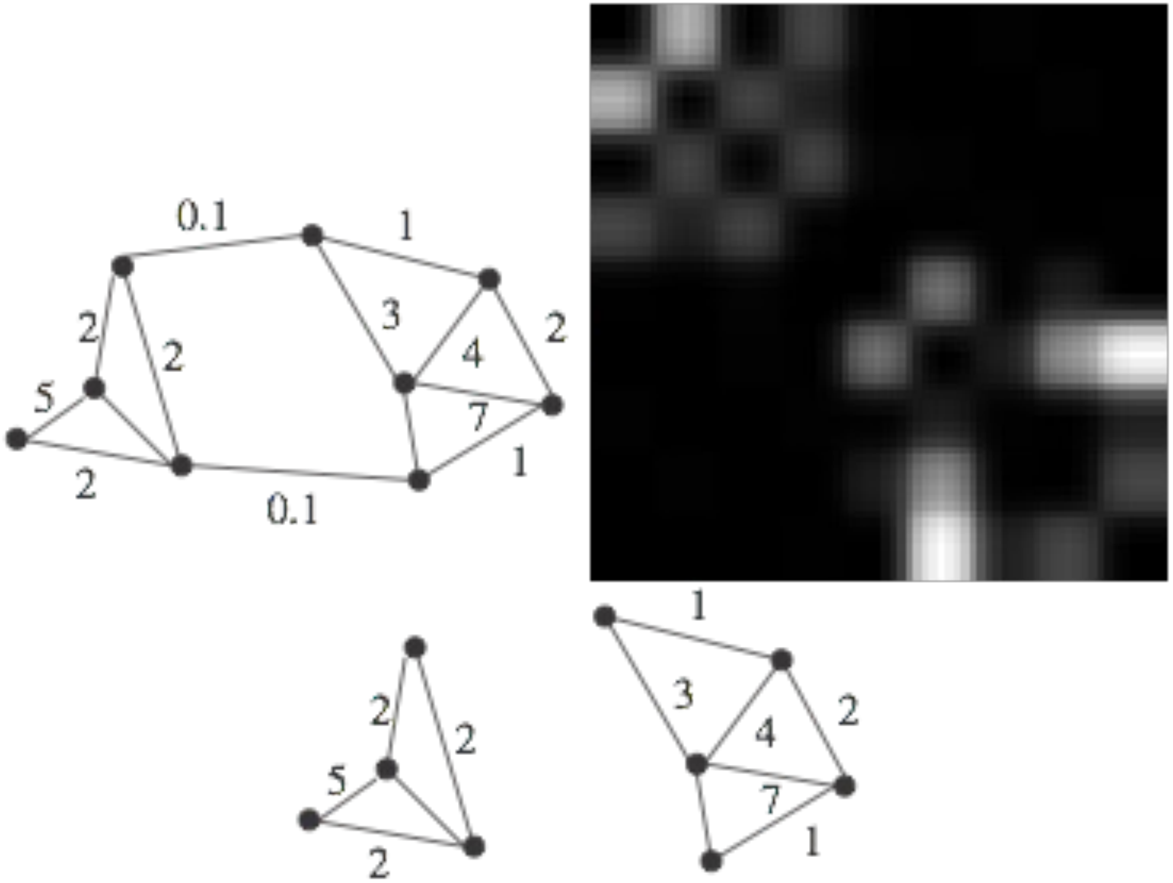
Pairwise affinities define a graph





- A graph is a set of vertices V and edge E that connects various pairs of vertices. Each edge can be represented as a pair of vertices.
- A directed graph is one in which edges (a,b) and (b,a) are distinct.
- An undirected graph is one in which no distinction is drawn between edges (a,b) and (b,a).

Pairwise affinities define a graph



Eigenvectors and segmentation

What is a good cluster?

The element associated with each cluster should have large values connecting one another in the affinity matrix.

$$\mathbf{w}_n^T \mathcal{A} \mathbf{w}_n \quad \text{with} \quad w_n[i] := \text{association of element i to cluster n}$$

A scaling factor needs to be introduced

$$\mathbf{w}_n^T \mathbf{w}_n = 1$$

Eigenvectors and segmentation

We want to find good clusters by maximising:

$$\mathbf{w}_n^T \mathcal{A} \mathbf{w}_n \quad \text{subject to:} \quad \mathbf{w}_n^T \mathbf{w}_n = 1$$

This is equivalent to finding the eigenvalues of A:

$$\mathcal{A}\mathbf{w}_n = \lambda \mathbf{w}_n$$

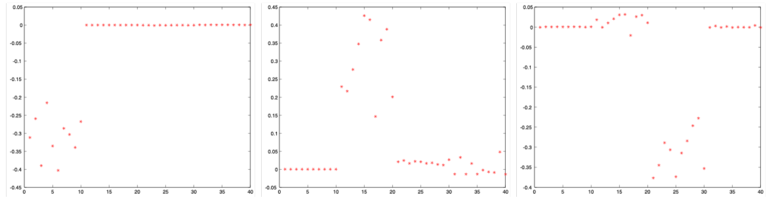
Clustering by graph eigenvalues

Construct the affinity matrix

Compute the eigenvalues and eigenvectors of the affinity matrix

Until there are sufficient clusters:

- Take the eigenvector to the largest unprocessed eigenvalue
- Zero all the components that have already been clustered
- Determine the components of the cluster by thresholding



Segmentation by graph-theoretic clustering

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- The graph will be cut into connected components that have relatively large interior weights by cutting edges with low weights.

Affinity measures

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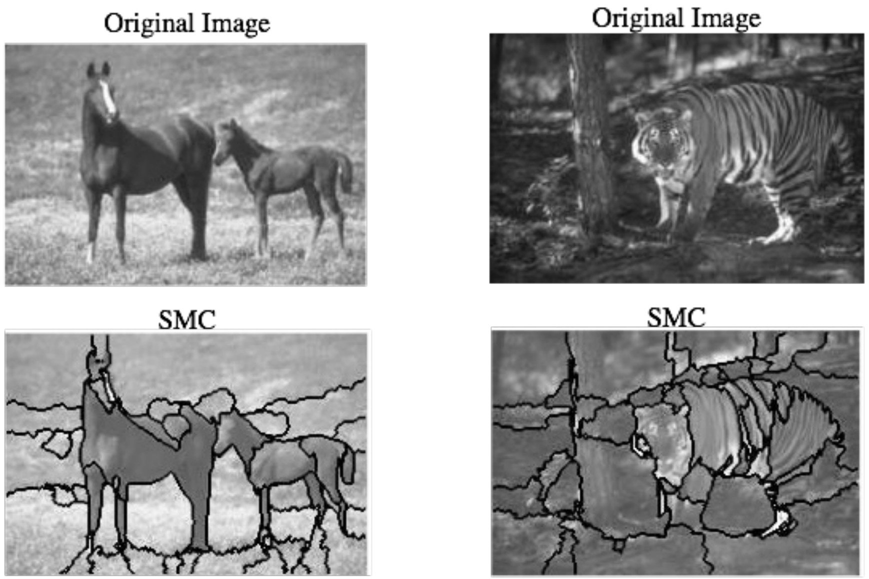
Example

This approach has a bias for partitioning the image into a small set of segments



F.J. Estrada, A.D. Jepson, and C. Chennubhotla. Spectral embedding and min-cut for image segmentation. In *British Machine Vision Conference*, 2004

Normalized cuts

An alternative approach is to cut the graph V into connected components such that the cut is a small fraction of the total affinity within each group.

Cutting groups A and B

$$\mathrm{cut}(A, B) = \sum_{v \in A} \sum_{w \in B} \mathrm{aff}(v \to w)$$

Association with group A

$$\mathrm{assoc}(A, V) = \sum_{v \in V} \sum_{w \in A} \mathrm{aff}(v \to w)$$

J. Shi and J. Malik. Normalized cuts and image segmentation. *IEEE Trans. Pattern Anal. and Machine Intell.*, 22(8):888–905, 2000.

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Total connection of all nodes in A to other nodes in the graph

$$\mathrm{assoc}(A,V) = \sum_{v \in V} \sum_{w \in A} \mathrm{aff}(v \to w)$$

J. Shi and J. Malik. Normalized cuts and image segmentation. *IEEE Trans. Pattern Anal. and Machine Intell.*, 22(8):888–905, 2000.

Normalized cuts

The normalised cut criterion is formulated as

$$\frac{\mathrm{cut}(A,B)}{\mathrm{assoc}(A,V)} + \frac{\mathrm{cut}(A,B)}{\mathrm{assoc}(B,V)}$$

The cut is small if the cut separates two components that have few edges of low weight between them and many internal edges of high weight.

Don't look at the value of the total edge weight connecting two partitions, measure the cut cost as a fraction of the total edge connections to all the notes in the graph.

Normalized cuts

The degree matrix is given as:

$$\mathcal{D} \quad \text{with} \quad D_{ii} = \sum_i \mathcal{A}_{i,j}$$

The normalised cut satisfies:

$$(\mathcal{D} - \mathcal{A})\mathbf{y} = \lambda \mathcal{D} \mathbf{y}$$

The normalized cut grouping algorithm

- Set up the weighted graph G = (V, E)

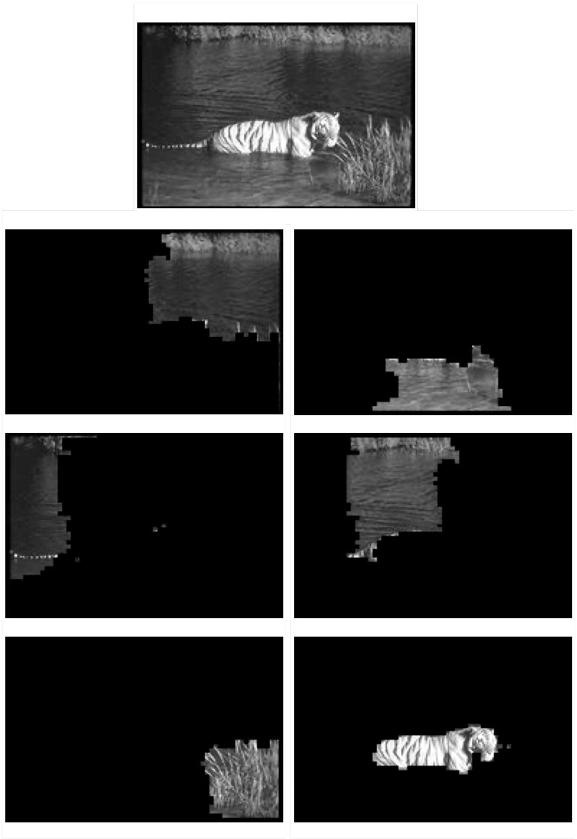
- Solve the following eigenvalue problem with the smallest eigenvalue

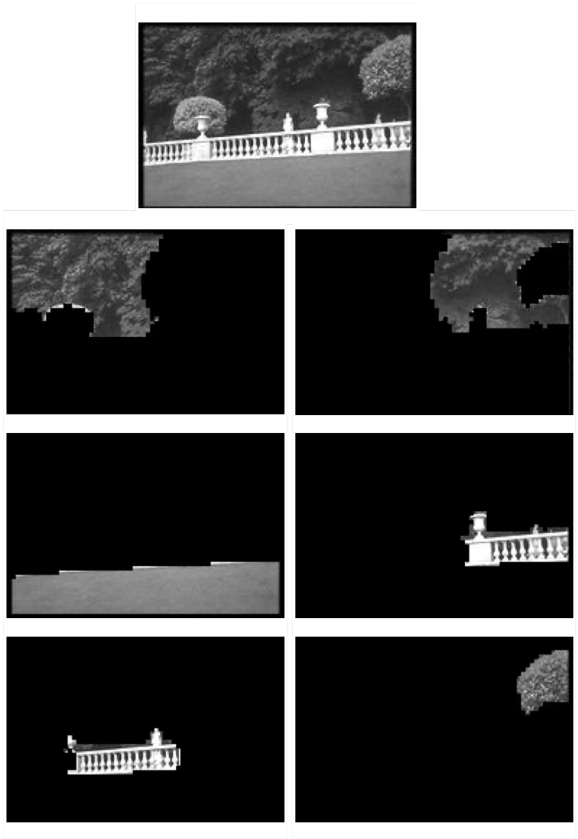
$$(\mathcal{D} - \mathcal{A})\mathbf{y} = \lambda \mathcal{D} \mathbf{y}$$

- Use the eigenvalue with the second smallest eigenvalue to bipartition the graph

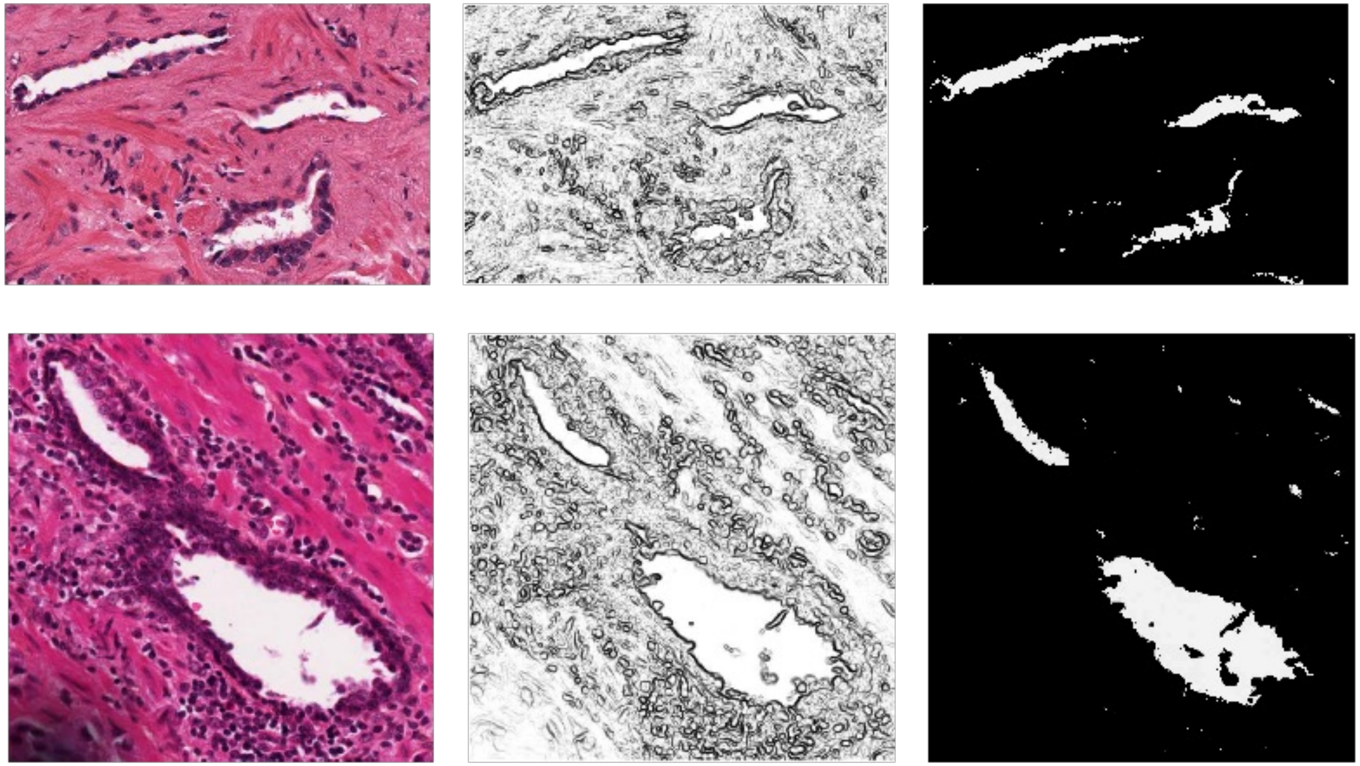
- Decide if the current partition should be subdivided and recursively partition the segmented parts of necessary

Normalized cuts - example





Normalized cuts - example



J. Xu, A. Janowzyk, S. Chandran, and A. Madabhushi, A Weighted Mean Shift, Normalized Cuts Initiated Gradient Based Geodesic Active Contour, SPIE 2010

Normalised cuts – practical considerations

Pairwise affinity matrices can be very large.

The corresponding eigenvalue problem can be computationally very expensive.

Superpixels can be used effectively to reduce the complexity of the problem

500 pixels x 500 pixels

↓

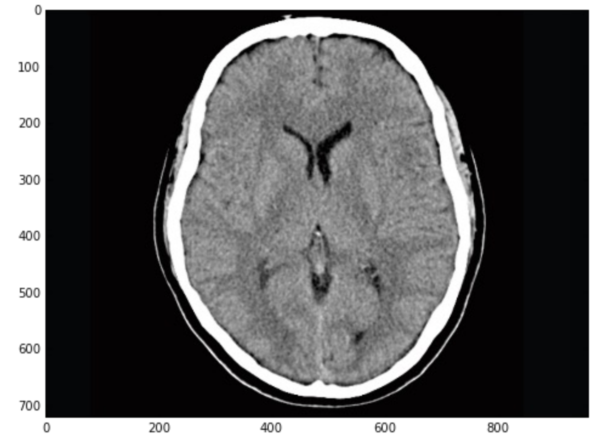
A – 250k x 250k

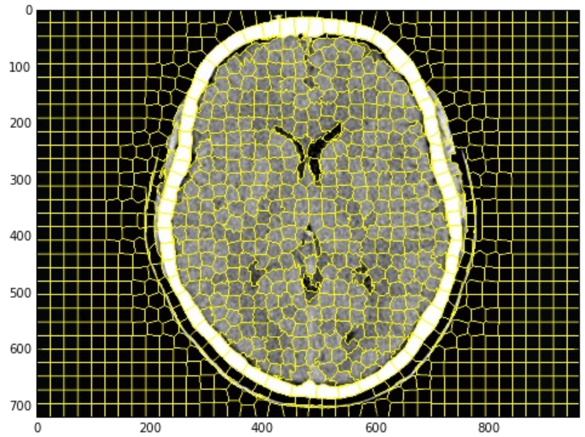
2000 pixels x 2000 pixels

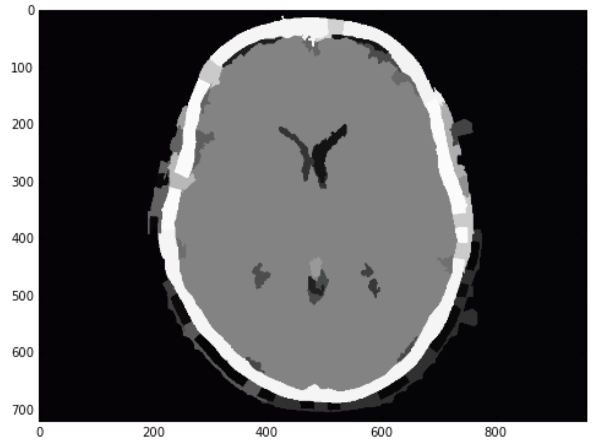
↓

A – 4M x 4M

Normalised cut on superpixels: example







N = 1200
Compactness = 20.0

Interactive graph cuts

Interactive segmentations play a big role in medical imaging. Fast and reliable segmentation methods can be used in the clinical workflow.

Interactive graph cuts present a very powerful approach to this problem.

Every pixel can be assigned to either foreground or background. The assignments are captured using a binary vector:

$$A = (A_1, \ldots, A_p, \ldots, A_{|P|}) \quad \text{with} \quad A_p \in 0, 1$$

Interactive graph cuts

Labelling problems can be formulated in terms of an energy minimization:

$$E(A) = \sum_{p \in P} D_p(A) + \lambda \sum_{p,q} V_{p,q}(A) \quad \text{with} \quad \lambda \geq 0$$

Penalty for
Pixel labelling

Smoothing term
pixel interactions

Energy function

As we are interested in estimating the optimal boundary the energy term becomes:

$$E(A) = \underbrace{\sum_{p \in P} R_p(A)}_{\text{Regional term}} + \underbrace{\lambda \sum_{p,q} B_{p,q}\delta(A)}_{\text{Boundary term}} \quad \text{with} \quad \lambda \geq 0$$

Regional term

$$E(A) = \underbrace{\sum_{p \in P} R_p(A)}_{\text{Regional term}} + \lambda \sum_{p,q} B_{p,q} \delta(A) \quad \text{with} \quad \lambda \geq 0$$

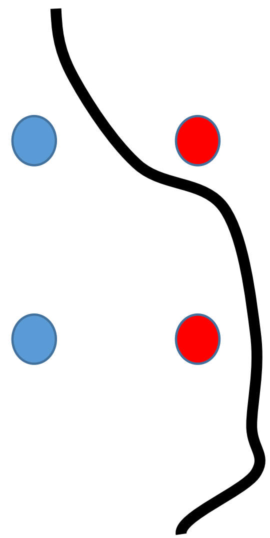
Penalize pixel label based on local properties

$$R_p(obj) = -\log P(I(p)|\mathcal{O}) \qquad R_p(bkq) = -\log P(I(p)|\mathcal{B})$$

Boundary term

$$E(A) = \sum_{p \in P} R_p(A) + \lambda \sum_{p,q} B_{p,q}\delta(A) \quad \text{with} \quad \lambda \geq 0$$

Boundary term

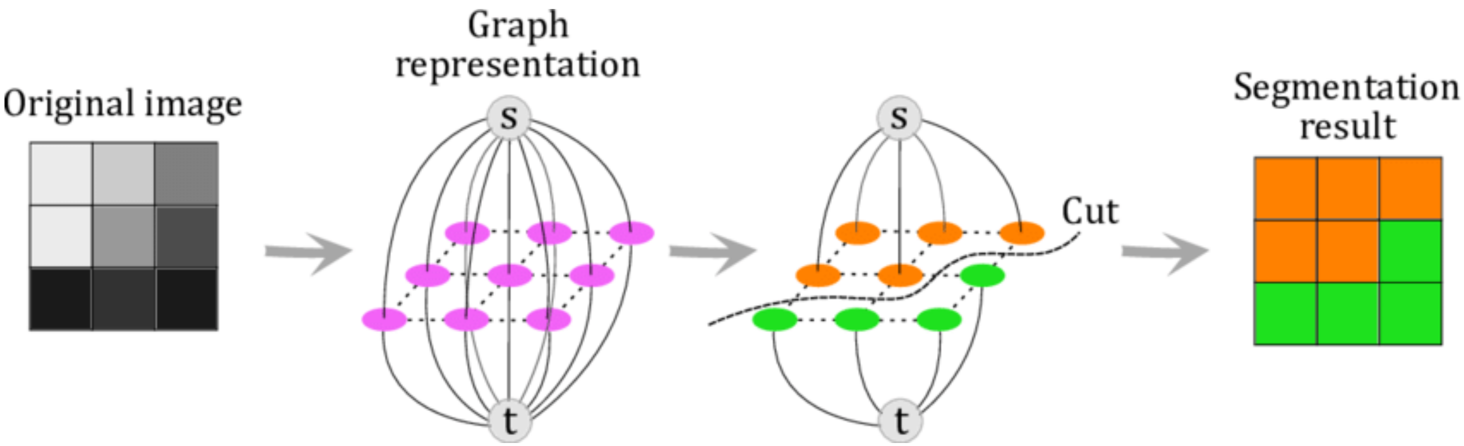


Penalize dissimilar neighbours

$$B_{p,q} \approx exp\left(-\frac{(I(p) - I(q))^2}{2\sigma^2}\right) \cdot \frac{1}{dist(p,q)}$$

Graph cuts

The graph cut can be solved by maximizing the flow through the graph



If the segmentation would be known:

- Terminal links (S->p) would be 1 of p in the foreground, otherwise 0
- Terminal links (p->T) would be 1 if p in the background, otherwise 0
- Neighbourhood links would be 1 of p,q are on a boundary, otherwise 0

Graph cuts

The probabilities defined in the cost function are now used to specify the cost for terminal links and neighbourhood links:

$$R_p(obj) = -\log P(I(p)|\mathcal{O}) \qquad R_p(bkq) = -\log P(I(p)|\mathcal{B})$$

$$B_{p,q} \approx exp\left(-\frac{(I(p)-I(q))^2}{2\sigma^2}\right) \cdot \frac{1}{dist(p,q)}$$

The iterative Floyd-Fulkerson algorithm can then be used to compute the maximum flow.

Interactive graph cuts - example

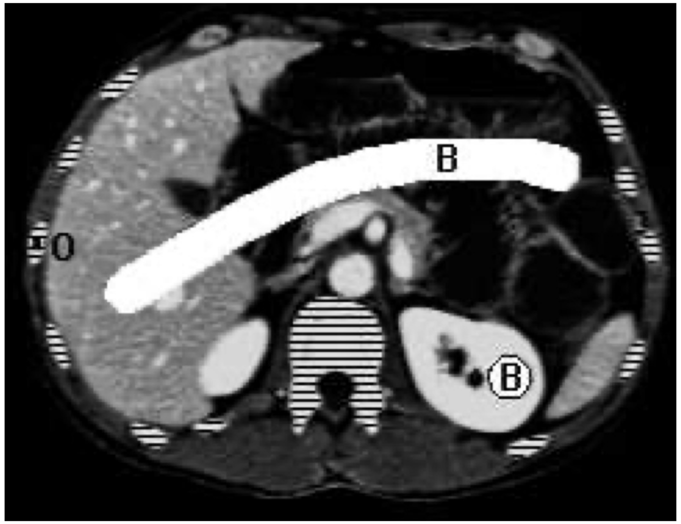


Figure 5. Bone removal in a CT image. Bone segments are marked by horizontal lines.

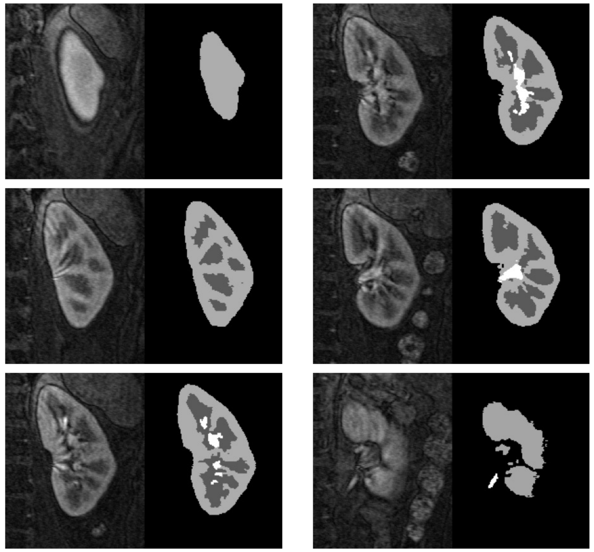


Figure 6. Kidney in a 3D MRI angio data.

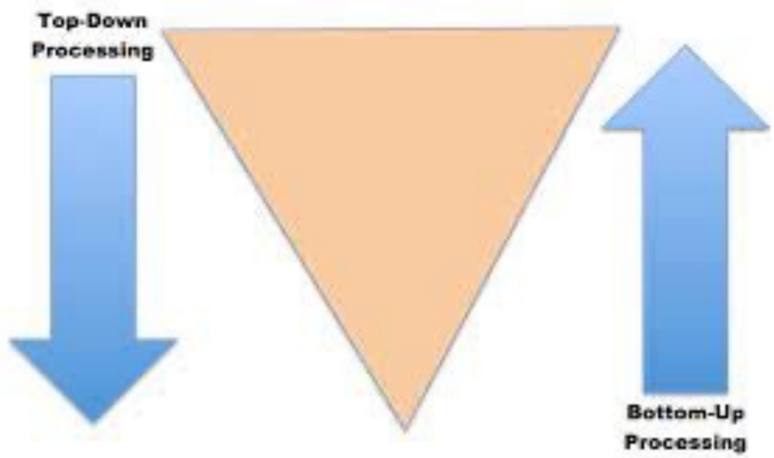
Boykov, Y.Y. and Jolly, M.P., 2001, July. Interactive graph cuts for optimal boundary & region segmentation of objects in ND images. In *Proceedings eighth IEEE international conference on computer vision. ICCV 2001* (Vol. 1, pp. 105-112). IEEE.

bottom-up vs top-down segmentation

- Inspired by Marr's vision we have attempt to define segments based on elementary sensory inputs (i.e. pixels)

- Then we group them based on perceptual or mathematical criteria

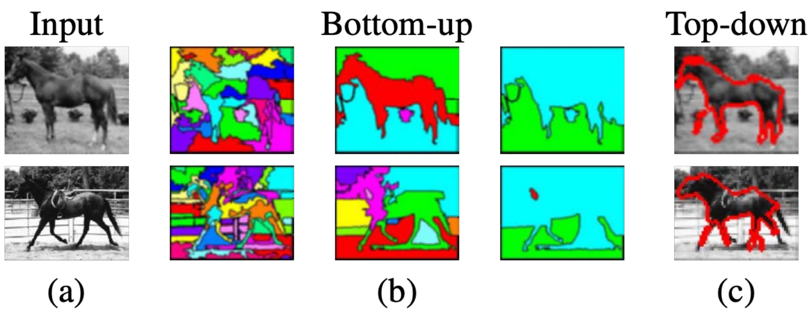
- Bottom-up approaches ignore higher level knowledge and prior information



Top-down vs bottom-up

- Bottom-up: guided by discontinuities the foreground object is split into multiple regions that merge with the background

- Top-down: a more meaningful approximation of the foreground object.



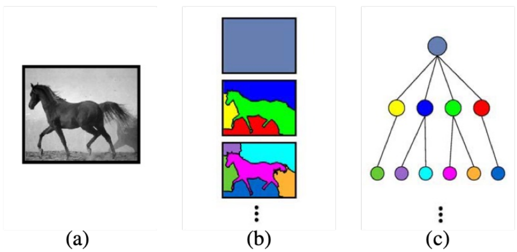


Figure 3: The bottom-up segmentation tree. (a.) Input image. (b.) Bottom-up segmentation of (a) at multiple scales. (c.) Segmentation tree. The bottom-up segments are organized in a tree structure. (The colors of the segments in (b) match those of the corresponding nodes in (c).)

Top-down approach

(a) (b) (c) (d) (e)

Figure 2: Overview of the top-down approach. (a.) Input image. Fragments stored in memory (b) are used to detect and cover the object (c).(d.) The figure-ground labeling of the covering fragments is used to label the image pixels as figure (blue region) or background. (e.) The boundary defined by the figure-ground segmentation (red contour).

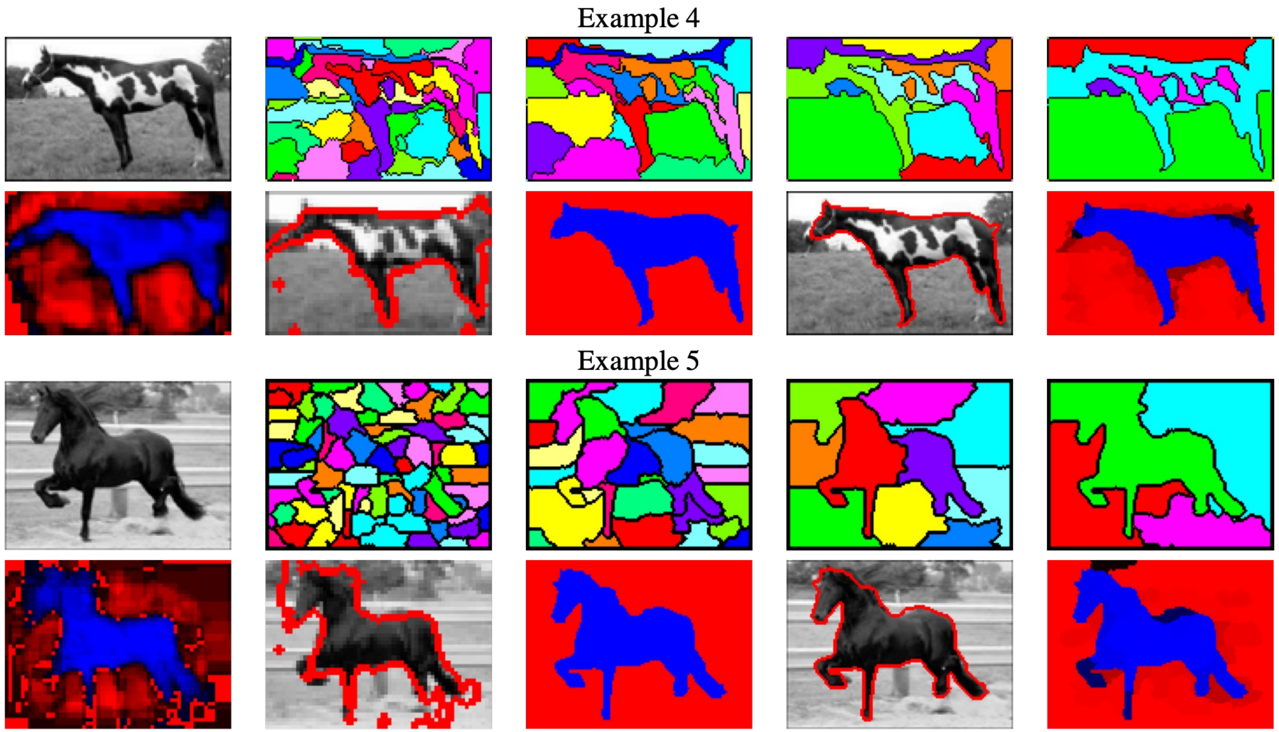
cost function

$$P(\bar{s}) = \sum_i t_i(s_i) + b_i(s_i, \bar{s_i})$$

Top-down cost depends on a classifier

Bottom-up term depends on local saliency

Examples



Validation of segmentation

We typically compare one or more manual (expert) segmentations with the automated segmentation method.

We need to:

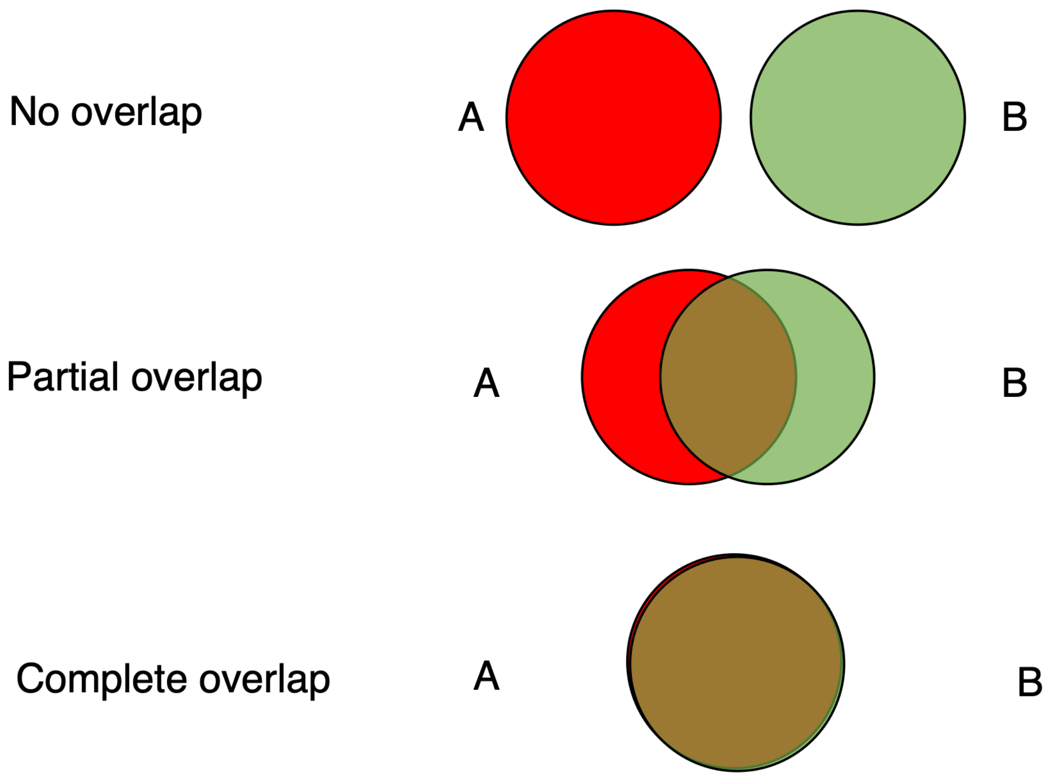
- Chose a validation data type (validation object)
- Define a measure of segmentation quality or a segmentation metric

We can then test how segmentation parameters (threshold, no. of iterations,...) affect the result

- This helps us to fine-tune the segmentation parameters for a given application

Segmentation metrics

Spatial overlap of target segmentations A and B



Region metrics

Dice Similarity Index (DSC or SI):

- Ratio of number of pixels in the intersection of two volumes, to the mean volume (this is sometimes called the “mean overlap”, MO)

$$DSC = \frac{2(A \cap B)}{A + B}$$

Tanimoto coefficient (TC) or Jacquard index:

- Also called the “union overlap”

$$TC = \frac{A \cap B}{A \cup B}$$

Measures are related:

- Same at extremes (0,1) DSC>TC in between (hence DSC often favoured...!)

$$DSC = \frac{2TC}{TC + 1}$$

Validation of segmentation: region metrics

Target overlap (TO)

- Measure of sensitivity
- Intersection of two volumes divided by the reference volume A:

$$TO = \frac{A \cap B}{A}$$

False negative and false positive errors:

$$FN = \frac{A \setminus B}{A} \qquad FP = \frac{B \setminus A}{B}$$

Validation of segmentation: region metrics

Multiple label overlaps:

- When comparing multiple labels simultaneously, the most straightforward approach would be to average any of the overlap measures across all labels.

- Example: DICE:

$$DSC_{avg} = \frac{1}{\# labels} \sum_{labels} \frac{2(A \cap B)}{A + B}$$

- However, this does not take into account overlapping labels (e.g. as may be caused by some errors in a deformation field, or independent structure segmentations)

Validation of segmentation: boundary metrics

- Region metrics* do not account for local variations in shape and boundaries.
- Boundary metrics can complement region metrics
- Very localized errors can be detected
- Can perform distance measurements between expert delineation and segmentation result
- Can use mean, RMS, maximum, median distance ...

Validation of segmentation: Hausdorff Distance

Measures worst match between 2 curves

- Given two curves, X={x1,..,xn}, Y={y1,..ym}, calculate closest distance from each x_i to Y as:

$$d(\mathbf{x}_i,\mathbf{Y}) = \min_{j=1}^{m} \left\| \mathbf{y}_j - \mathbf{x}_i \right\|$$

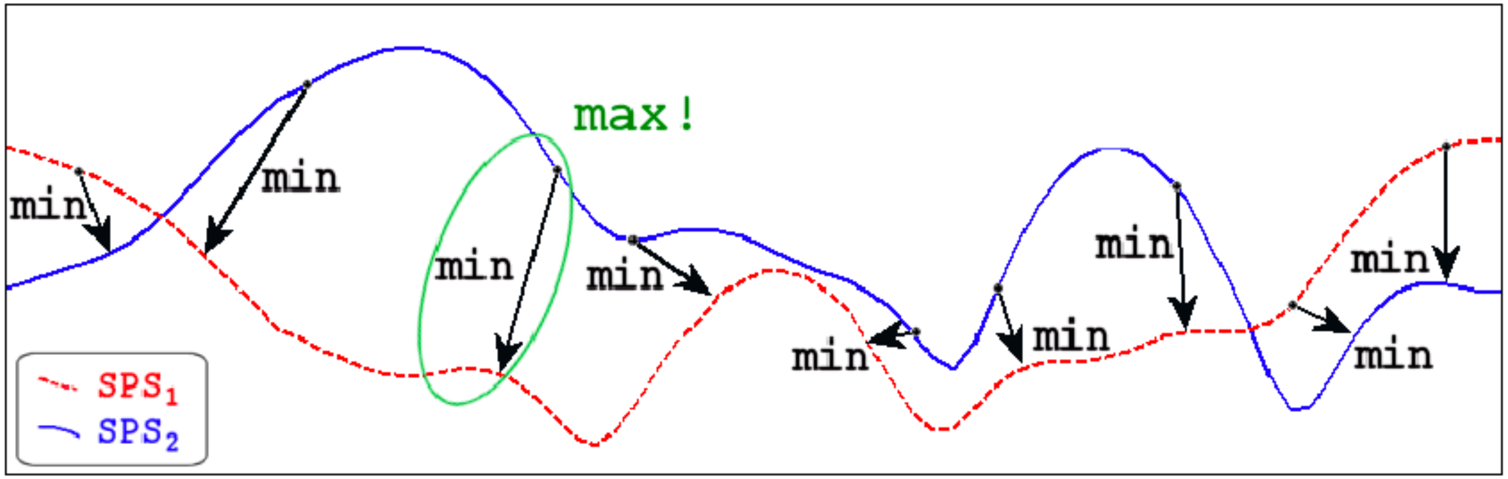
- The Hausdorff distance is defined as the maximum of all minimum distances of x_i to Y_i and hence measures the disparity of X and Y:

$$HD = \max\left(\left\{ d(\mathbf{x}_i,\mathbf{Y}) \right\}\right)$$

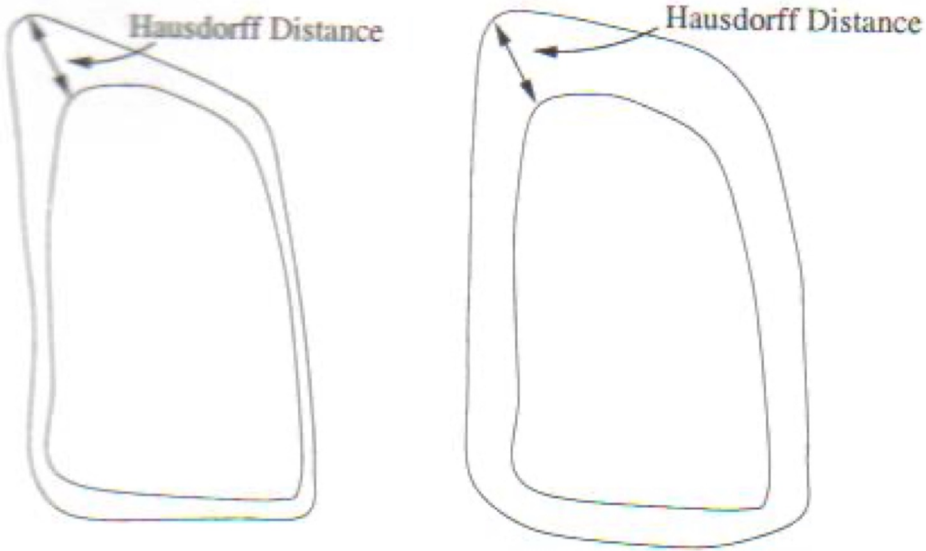
- A symmetric Hausdorff metric exists, which also measures the closest distances from y_i to X, and the returns the maximum distance

$$HD_{symm}(X,Y) = \max\left(\max\left(\left\{ d(\mathbf{x}_i,\mathbf{Y}) \right\}\right) \max\left(\left\{ d(\mathbf{y}_j,\mathbf{X}) \right\}\right)\right)$$

Hausdorff distance - example



Validation of segmentation: region vs boundary metrics



- Overlap is high (left), and low (right)
- Hausdorff distance is the same, but curves do not have the same (global) average distance

Questions

- Why can the naïve graph based clustering result in very small image regions?
- What are the main differences between bottom-up and top-down vision approaches?

Additional references

- An Analysis of Normalized Cuts and Image Segmentation [web]
- Metrics to Evaluate your Semantic Segmentation Model [web]
- Maier-Hein, L. and Menze, B., 2022. Metrics reloaded: Pitfalls and recommendations for image analysis validation. *arXiv. org*, (2206.01653)

List of Python notebooks

biomedia_seg_00_label_image	Working with label images and calculating measurements of a given segment
biomedia_seg_01_otsu	Calculating optimal thresholds using Otsu
biomedia_seg_02_connected_comp	Obtaining a label image using connected components
biomedia_seg_03_watershed	Determining region boundaries using watershed segmentation
biomedia_seg_04_kmeans	Applying clustering for image segmentation
biomedia_seg_05_superpixels	Application of SLIC to a histology image
biomedia_seg_06_normalised_cuts	Combining SLIC with the normalised cut criterion

Please refer to the additional documentation on github